Ritabrata Chakraborty

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Education

University of Pennsylvania — **Pennsylvania, USA**
M.S.E. in Robotics, ***GPA: –/4*** — *Aug '26 – May '28*

Birla Institute of Technology and Science, Pilani (BITS Pilani) — **Rajasthan, India**
B.E. in Mechanical Engineering — Minor in Data Science, ***CGPA: 8.57/10*** — *Oct '22 – May '26*

- **Institute Merit-cum-Need Scholarship** — Consecutive Recipient (Top 8% of Students)

Publications & Peer Reviews

Journal Publications

- **R. Chakraborty** and K. Kishore, ***"Navigating Without Satellites: A Critical Survey of GNSS-Denied UAV Navigation."*** Submitted to **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.
- **R. Chakraborty** and K. Kishore, ***"Multi-Critic Off-Policy Reinforcement Learning for Quadrotor Velocity Control in Unknown Indoor Environments."*** Manuscript in preparation for **IEEE Transactions on Artificial Intelligence (T-AI)**, 2026.
- **R. Chakraborty**, ***"A Safe Harbour in a Cut-Throat World? General and Engineering-Specific Primal World Beliefs Among Indian Engineering Undergraduates."*** Submitted to **Journal of Social and Educational Research**, 2026.

Conference Publications

- Y. Wang, H. He, Y. Cao, J. Liang, **R. Chakraborty**, and G. Sartoretti, ***"CogniPlan: Uncertainty-Guided Path Planning with Conditional Generative Layout Prediction."*** Accepted at **Conference on Robot Learning (CoRL)**, 2025; published in **Proceedings of Machine Learning Research (PMLR)**.
- **R. Chakraborty**, T. Mian, and P. Kundu, ***"An Efficient Approach for Synthetic Data Generation and Fault Diagnosis for Rotating Machinery."*** Accepted at **Prognostics and System Health Management Conference (PHM)**, 2025; published in **IET Conference Proceedings**.
- K. Kishore and **R. Chakraborty**, ***"Path Planning of Low-Altitude UAV for Tree Canopy Tracking and Orchard Monitoring."*** Publication awaiting intellectual property clearance, 2025.

Poster Presentations

- **R. Chakraborty**, ***"High-Velocity Impact Testing and Analysis of Layered Metal-Composite Shields Using Electromagnetic Railgun."*** Presented as poster at **Pulsed Power Science, Technology and Application (PPSTA)**, 2024.

Peer Reviews

- ***"GS-SDE: 3D Gaussian Splatting-based Adaptive Scene Modeling in Semi-Dynamic Environments."*** Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- ***"RoSe6D: Role-Separated Structure-Aware Guidance for RGB-Based 6D Object Pose Estimation."*** Peer review completed for **IEEE Transactions on Automation Science and Engineering (T-ASE)**, 2026.
- ***"Limits of Wearable Gait Analysis for Age Classification in Young Adults: Anthropometry Over Age."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"Generative AI and Computer Vision Based Multi-Assessment System for Early Prediction and Diagnosis of Autism Spectrum Disorder (ASD)."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"AquaNode: Design and Implementation of a Low-Cost Automated Water Tank Monitoring and Valve Control System Using Arduino and Ultrasonic Sensing."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"A research paper on IoT-Based Smart Irrigation Systems."*** Peer review completed for **IEEE India Council International Subsections Conference (INDISCON)**, 2026.
- ***"Hybrid AI-Based Autonomous Drone Swarm Navigation."*** Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- ***"Autonomous Robot Navigation for Smart Indoor using A* and Deep Q-Network Model."*** Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.
- ***"Deep Learning Driven Robot Simulation for Autonomous Navigation, Crop Health and Pesticide Spraying."*** Peer review completed for **IEEE Gujarat Section Conference (GSC)**, 2026.

Experience – Robotics & Computer Vision

Vision-Language Grounded Multi-Robot Coordination and Navigation

Supervisor: Dr. Avinash Gautam, Associate Professor, BITS Pilani — *May '25 – Aug '25*

- Surveyed literature on vision-language grounded multi-robot coordination and VLMaps-based spatial representations.
- Reviewed decentralized navigation strategies fusing 3D visual-language maps with LLM-driven instruction parsing.

Off-Policy Multi-Critic RL with Reward Decomposition for Indoor Exploration

Supervisor: Dr. Kaushal Kishore, Principal Scientist, CSIR-CEERI — *Jan '26 – May '26*

- Integrated three specialized critics for forward progress, lateral avoidance, and angular alignment with distinct reward signals.
- Deployed multi-critic TD3 with prioritized replay achieving 94.2% success rate and 22% lower collision rate versus baseline.

Vision-Attention-Driven Autonomous Navigation with Semantic Understanding

Supervisor: Dr. Guillaume Sartoretti, Assistant Professor, MARMoT Lab, NUS — *Aug '25 – Dec '25*

- Enhanced CogniPlan with cross-attention between frontier and node embeddings to enrich node representations.
- Incorporated Visual Navigation Transformer (ViNT) to capture semantic context for adaptive exploration strategies.

Design and Development of Smart Automated Field Hockey Ball Launcher

Supervisor: Dr. Mani Shankar Dasgupta, Associate Professor, BITS Pilani — *Apr '24 – Oct '25*

Physics-Based Launcher Mechanical Design and FEM Analysis

- Fabricated programmable launcher achieving 150 km/h using CFRP composite and stainless steel components.
- Performed FEM analysis and CAD using Fusion 360, achieving safety factor 2.07 at 496 RPM.

Computer Vision Pipeline for 3D Trajectory Reconstruction and Shot Classification

- Devised 3D ball localization from monocular videos using diameter-based depth and PnLCalib (0.76m RMSE).
- Constructed TCN-attention classifier fusing trajectory and statistical features, achieving 96.4% classification accuracy.

Uncertainty-Guided Path Planning via Conditional Layout Prediction

Supervisor: Dr. Guillaume Sartoretti, Assistant Professor, MARMoT Lab, NUS — *Mar '25 – Aug '25*

- Adapted multiple image-based exploration planners into ROS pipelines for evaluation in the CMU Exploration Environment.
- Supported the development and benchmarking of CogniPlan, achieving up to 17.7% shorter exploration paths and 3.9% improved navigation efficiency over state-of-the-art baselines across 100+ unseen maps.

Monocular Vision-Based UAV Navigation for Orchard Monitoring

Supervisor: Dr. Kaushal Kishore, Principal Scientist, CSIR-CEERI — *Jan '24 – Feb '25*

- Engineered a UAV-based orchard monitoring system using YOLOv11 (Box mAP50: 95.5%, Mask mAP50: 96.5%).
- Accelerated YOLOv11 inference for real-time onboard navigation by exporting to ONNX and building an FP16 TensorRT engine on an NVIDIA Jetson Orin (17 FPS, 102.3 ms latency).

- Implemented B-spline trajectory logic and coded custom yaw-roll controller to minimize drift under wind.

Image-Based Visual Servoing for Automated Radar Control and UAV Tracking
Supervisor: Dr. Rishi Verma, Head (ESDS), PPEMD, BARC *Jun '24 – Jul '24*
- Established automated radar alignment using YOLOv8 and DeepSORT, improving tracking recall to 91% and speed by 13%.
- Interfaced a PLC-based actuation system enabling continuous UAV lock-in within $\pm 5^{\circ}$ angular error.

Experience – Generative AI & Predictive Maintenance

Synthetic Data Generation for Bearing Fault Diagnosis Under Data Scarcity
Supervisor: Dr. Pradeep Kundu, Assistant Professor, KU Leuven *Sep '24 – May '26*

Auxiliary Classifier WGAN-GP for Time-Series Sensor Data Generation
- Built ACWGAN-GP with TCN discriminator for minority fault class augmentation reaching ∼100% classification accuracy.
- Assessed synthetic data using PCC, Cosine Similarity, MMD, and KL Divergence on FFT spectra.

Physics-Informed Generative Modeling for Bearing Fault Diagnosis Under Data Scarcity
- Formulated physics-informed cVAE-GAN and cVQ-GAN frameworks isolating latent-space effects in CWT scalogram synthesis.
- Validated LiteFormer classification gains across Density, Coverage, and CMMD metrics.

Experience – Mechanical Simulation & Behavioral Research

Finite Element Analysis of Running Track Surfaces
Supervisor: Dr. Mani Shankar Dasgupta, Associate Professor, BITS Pilani *Apr '24 – Sep '25*
- Modeled surface materials using hyperelastic formulations, evaluating restitution across 3 sports disciplines.
- Proposed optimized track designs improving durability by 23% based on ANSYS fatigue analysis.

Primal World Beliefs in Engineering Students
Supervisor: Dr. Rajneesh Choubisa, BITS Pilani *Aug '24 – Dec '24*
- Conducted a primal world beliefs study surveying engineering undergraduates on core worldview dimensions.
- Replicated established population-level findings, validating trends against the original general-population study.

Impact Testing and Analysis of Metal-Composite Shields Using Electromagnetic Railguns
Supervisor: Dr. Rishi Verma, Head (ESDS), PPEMD, BARC *Jun '24 – Jul '24*
- Characterized 3 material configurations via the Johnson-Cook model for high-strain-rate impact response.
- Verified simulation results through railgun tests at 3.2 MJ energy, achieving <7% deviation from FEM predictions.

Industry Experience

ML Intern | Uber (AI Solutions)
Supervisor: Mr. Siddarth Malreddy, Tech Lead Manager & Mr. Ishan Nigam, Senior ML Engineer, Uber *Jul '25 – Dec '25*
- Streamlined human-in-the-loop automated object tracking in uLabel for RGB/IR videos.
- Flagged object annotation anomalies via XGBoost based on spatial and temporal consistency (F1: 82%).
- Screened object-association errors with a TrackTrack-based cross-frame consistency check (F1: 88%).
- Served the tracking pipeline as a CPU-only REST API on GCP via Uvicorn, processing video chunks in parallel with result caching to cut latency.

Subject Matter Expert | Knowledge Cultivator
Self-supervised *Nov '23 – Feb '24*
- Resolved numerical problems for student clients using MATLAB, CAD, and Python.
- Authored academic content across diverse research topics.

Projects

Vision-Guided Robotic Arm for Simulated Oral Cancer Screening *Aug '25 – Dec '25*
- Simulated a KUKA KR 150 R3100 arm in MATLAB/Simulink for end-effector guidance inside a modeled oral cavity.
- Trained an EfficientNet-B0 model to detect cancerous regions, localizing the end effector through ORB-SLAM3 tracking.

Autonomous Drone Navigation | MathWorks Global Student Drone Challenge 2025 *Mar '25 – Apr '25*
- Designed vision-based control for Parrot Mambo drone using masking, ray-tracing, and closed-loop yaw control.
- Optimized speed control and zone-based auto-landing to reduce track completion time.

Air Quality Index Forecasting Using Time Series Models *Nov '24 – Nov '24*
- Benchmarked AR, MA, ARMA, ARIMA, and SARIMA models forecasting daily and weekly AQI for Jaipur and Kolkata.
- Diagnosed stationarity with ADF and KPSS tests, tuning lag orders via ACF/PACF and scoring on R^2, MAE, and RMSE.

3D Lunar Surface Modeling from DEM Data | ISRO Immersion Challenge 2024 *Jul '24 – Jul '24*
- Created lunar terrain models from TMC DEM data with curvature extraction, rendered live in UE5.
- Automated data import and heightmap transformation via a custom plugin, processing each tile in under 10s.

Cricket Team Optimization | AMEX Campus Challenge 2024 (Product Track) *Jun '24 – Jul '24*
- Processed and cleaned 100K+ datapoints to identify anomalies, outliers, and consistency trends.
- Generated optimal lineups using statistical heuristics, achieving a 14% gain over baseline strategies.

ExoMy Rover Navigation and UR3 Arm Motion Planning | ERC 2023 Remote *Apr '23 – Sep '23*
- Navigated ExoMy rover using ArUco detection, Ackermann steering, and spot turns for autonomous hazard avoidance.
- Calibrated UR3 arm with MoveIt and OMPL planner for collision-free manipulation (98% success).

Awards & Achievements

Finalist — Meta PyTorch OpenEnv Hackathon (India) *Apr '26*
3rd Place — MathWorks Global Drone Student Challenge 2025 *Mar '25*
Finalist — ISRO Immersion Challenge (AI for Space & Geospatial Innovation) *Jul '24*
Top 15 Overall, Top 5 in College — American Express Campus Challenge 2024 *Jul '24*
5th Place & Best Maintenance Award — European Rover Challenge 2023 (Remote Finals) *Sep '23*

Teaching

Teaching Assistant
CIS 4190/5190: Applied Machine Learning, University of Pennsylvania *Aug '26 – Dec '26*
CS F320: Foundations of Data Science, BITS Pilani *Jan '26 – May '26*
ME F218: Advanced Mechanics of Solids, BITS Pilani *Jan '25 – May '25*
ME F216: Materials Science and Engineering, BITS Pilani *Sep '24 – Dec '24*
- Assisted 100+ students in labs and tutorials, clarifying concepts and linking theory to practical applications.
- Analyzed assignments and facilitated faculty in delivering high-impact teaching sessions.

Leadership & Volunteering

Deputy Director of Academic Programming

Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania *Aug '26 – Present*

- Spearheaded GradFest, engaging 500+ graduate and professional students across Penn's schools.

Vice-President

South Asian Association (Rangoli), University of Pennsylvania *Aug '26 – Present*

- Co-lead planning and execution of cultural events and member engagement for Penn's South Asian student community.

President & Secretary

Mechanical Engineering Association (MEA), BITS Pilani *Jun '24 – May '26*

- Coordinated 10+ events and career sessions for 300+ students, facilitating technical exposure and alumni interaction.
- Managed production of 500+ merchandise items and led outreach, boosting student participation by 20%.

President & Tech Fest Coordinator

Indian Society of Heating, Refrigerating, and Air Conditioning Engineers (ISHRAE), BITS Pilani *Oct '24 – Jul '25*

- Led a team of 25+ members to organize 4+ technical workshops with HVAC industry experts.
- Hosted 3 competitions and networking events, engaging 200+ students in HVAC innovation and awareness.

Project Manager

Tinkerer's Lab (TL), BITS Pilani *May '24 – Jul '25*

- Supervised 5 interdisciplinary robotics teams (30+ members) on Micromouse and Hexapod projects.
- Oversaw lab resources, conducted weekly reviews, and mentored 50+ students in hands-on technical skills.

Treasurer

Bengali Cultural Association (Moruchhaya), BITS Pilani *May '24 – Jul '25*

- Allocated funds and maintained financial records for merchandise sales, cultural programs, and annual events.

Volunteer

Junoon, National Service Scheme (NSS), BITS Pilani *Sep '23 – Sep '23*

- Organized sports and recreational competitions for specially abled children from 25+ colleges.

Professional Activities

Graduate Student Member — IEEE *2026*

Competitive Examinations

GRE: 317 (Quantitative: 168, Verbal: 149, AWA: 3) *2025*
TOEFL iBT: 108 (Reading: 27, Listening: 30, Speaking: 23, Writing: 28) *2025*
JEE Advanced: All India Rank 14372 *2022*

Technical Skills & Coursework

Relevant Coursework: Machine Learning, Deep Learning, Machine Perception, Computer Vision, Foundations of Data Science, Applied Statistical Methods, Linear Algebra, Engineering Optimization, Differential Equations, Design of Machine Elements, Digital Twins
Programming Languages: Python, C++, C, C#, Java, Linux (Shell)
Robotics & Simulation: ROS (with Gazebo, Rviz), MAVROS, Navigation Stack, MoveIt, AirSim, Unity, Unreal Engine, MATLAB, Simulink, QGIS
AI & Data Science: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, OpenCV, Open3D, NumPy, Pandas, SciPy, StatsModels, Matplotlib, Seaborn, Weights & Biases (W&B), Jupyter, Excel
Cloud & MLOps: AWS (EC2, SageMaker), Azure (Foundry), Google Cloud Platform (GCP), REST APIs, Docker, ONNX, TensorRT, Git, GitHub Actions
Hardware & Embedded Systems: NVIDIA Jetson (Nano, Orin), Raspberry Pi, Arduino, IMUs, Stereo Camera, 3D LiDAR
CAD & Mechanical Simulation: ANSYS Mechanical, SolidWorks, Fusion 360
Leadership & Communication: Event Planning, Team Management, Budget Allocation, Mentorship, Cross-Functional Coordination, Public Speaking, Stakeholder Outreach, Technical Writing
Productivity Software: Microsoft Office (Excel, PowerPoint, Word), Google Workspace